	S.S.V.P.S's B. S. Deore Polytechnic, Dhule **Department of Computer Engineering** *Academic Year : 2025 – 2026*

Presentation on Capstone Project Execution (CPE–316004)

Topic	Blockchain Student Record System	
Presented by	1. PATIL OM HIRALAL	
	2. PATIL PUSHPAK GYANU	
	3. UPASANI DURGESH PRADIP	
	4. MARATHE LATIKESH SUBHASH	
Guided by	Mr. Niket Sharma	
Semester	**Sixth (K Scheme)**	**Date :**

Abstract

- This project presents a blockchain-inspired student record management system designed to ensure secure, tamper-proof storage of academic data. The system encrypts student profile information using SHA-256 and stores records as immutable blocks. Teachers verify student profiles through a structured workflow, while students can request changes before final approval. JWT authentication ensures secure access, and real-time dashboards provide verification analytics. The system improves transparency, security, and reliability in managing student academic records.

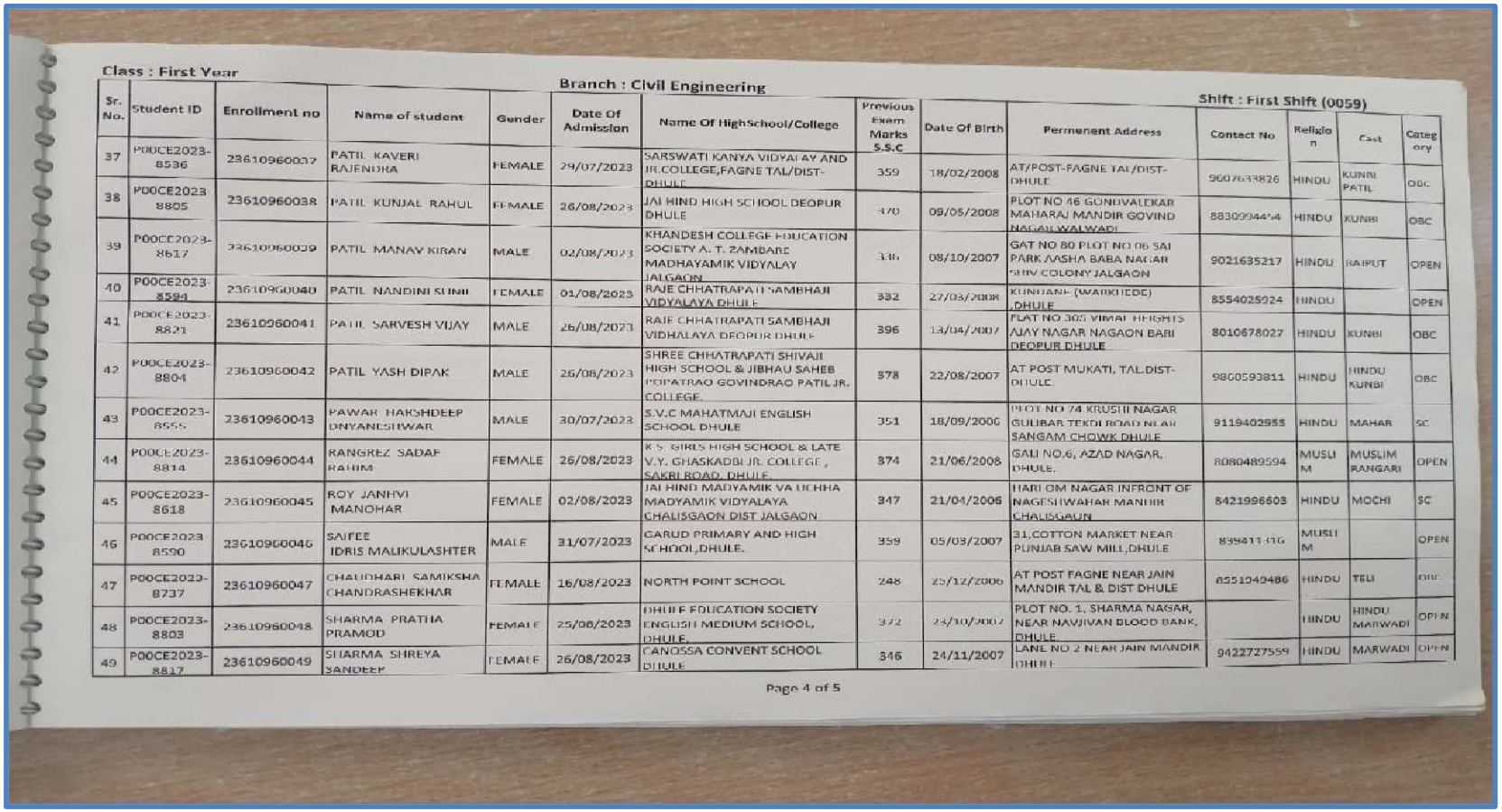
Content

1. Introduction. (background of the industry or used based problem / task)
2. Literature Survey.
3. Scope of the project.
4. Methodology.
5. Details of designs, working and processes.
6. Results and applications.
7. Conclusion and future scope.
8. References and Bibliography.

1. Introduction

- Student academic records such as Personal information like a Student Name, Enrollment number, Academic Performance, Contact and Guardian information. However, many institutions still use centralized databases and manual verification systems, which can lead to problems like fake certificates, data modification, and slow verification processes.
- In the traditional system, all data is stored in one central database, which creates risks such as hacking, unauthorized access, or data loss. To solve these issues, this project uses a blockchain-inspired approach.
- In this system, student data is securely stored and a unique hash is generated using the SHA-256 algorithm. This ensures that once the data is stored, it cannot be modified. During verification, the system compares hashes to confirm the authenticity of records, making the verification process faster, secure, and reliable.

2. Literature Review



Class : First Year

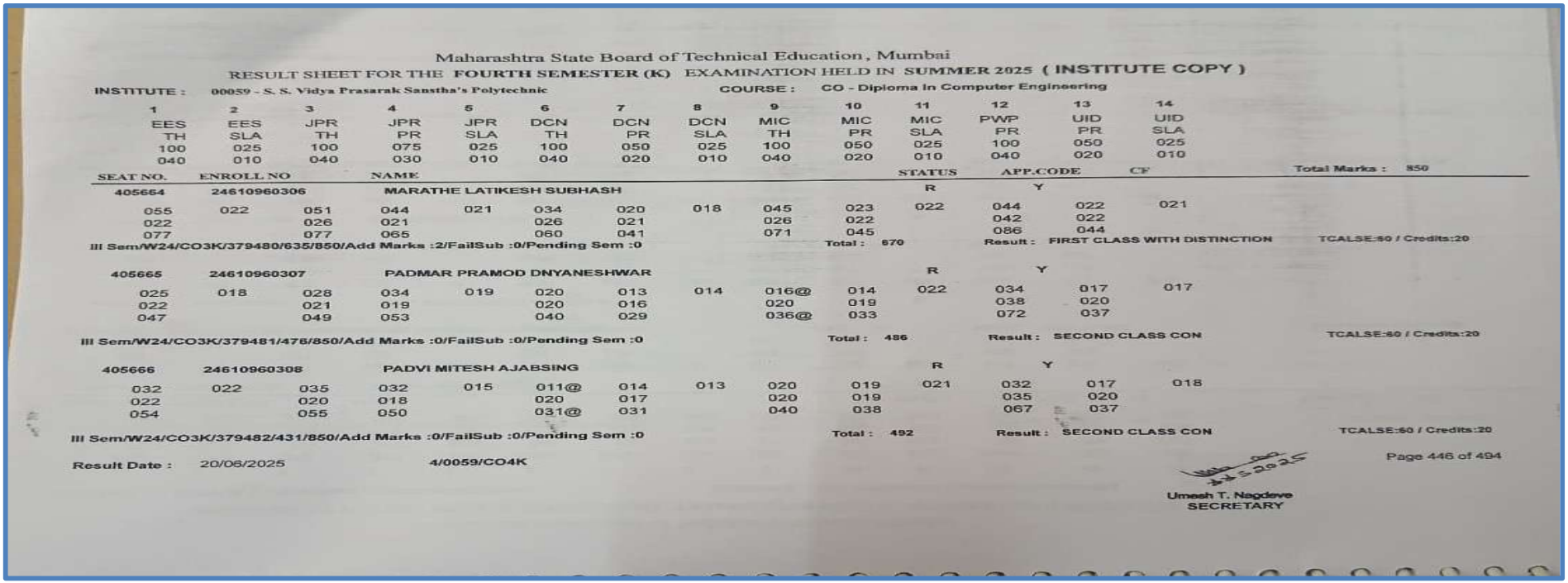
Branch : Civil Engineering

Shift : First Shift (0059)

Sr. No.	Student ID	Enrollment no	Name of student	Gender	Date Of Admission	Name Of HighSchool/College	Previous Exam Marks S.S.C	Date Of Birth	Permenent Address	Contact No	Religion	Cast	Category
37	P00CE2023-8536	23610960037	PATIL KAVERI RAJENDRA	FEMALE	29/07/2023	SARSWATI KANYA VIDYALAY AND JR.COLLEGE,FAGNE TAL/DIST-DHULE	359	18/02/2008	AT/POST-FAGNE TAL/DIST-DHULE	9007634826	HINDU	KUNBI PATIL	OBC
38	P00CE2023-8805	23610960038	PATIL KUNJAL RAHUL	FEMALE	26/08/2023	JAI HIND HIGH SCHOOL DEOPUR DHULE	470	09/05/2008	PLOT NO 46 GONDVALEKAR MAHARAJ MANDIR GOVIND NAGAR WALWADI	8830994454	HINDU	KUNBI	OBC
39	P00CE2023-8617	23610960039	PATIL MANAV KIRAN	MALE	02/08/2023	KHANDESH COLLEGE EDUCATION SOCIETY A. T. ZAMBARE MADHAYAMIK VIDYALAY JALGAON	336	08/10/2007	GAT NO 80 PLOT NO 06 SAI PARK AASHA BABA NAGAR SHIV COLONY JALGAON	9021635217	HINDU	RAJPUT	OPEN
40	P00CE2023-8594	23610960040	PATIL NANDINI SUNIL	FEMALE	01/08/2023	RAJE CHHATRAPATI SAMBHAJI VIDYALAYA DHULE	332	27/03/2008	KUNDANE (MARKETEDC) ,DHULE	8554025924	HINDU		OPEN
41	P00CE2023-8821	23610960041	PATIL SARVESH VIJAY	MALE	26/08/2023	RAJE CHHATRAPATI SAMBHAJI VIDHALAYA DEOPUR DHULE	396	13/04/2007	FLAT NO 305 VIMAL HEIGHTS AJAY NAGAR NAGAON BARI DEOPUR DHULE	8010678027	HINDU	KUNBI	OBC
42	P00CE2023-8804	23610960042	PATIL YASH DIPAK	MALE	26/08/2023	SHREE CHHATRAPATI SHIVAJI HIGH SCHOOL & JIBHAU SAHEB POPATRAO GOVINDRAO PATIL JR. COLLEGE.	378	22/08/2007	AT POST MUKATI, TAL.DIST-DHULE.	9860593811	HINDU	HINDU KUNBI	OBC
43	P00CE2023-8555	23610960043	PAWAR HAKSHDEEP DNYANESHWAR	MALE	30/07/2023	S.V.C MAHATMAJI ENGLISH SCHOOL DHULE	351	18/09/2006	PLOT NO 74 KRUSHI NAGAR GULIBAR TEKDI ROAD NEAR SANGAM CHOWK DHULE	9119402955	HINDU	MAHAR	SC
44	P00CE2023-8814	23610960044	RANGREZ SADAF RAHIM	FEMALE	26/08/2023	K.S. GIRLS HIGH SCHOOL & LATE V.Y. GHASKADBIR JR. COLLEGE , SAKRI ROAD, DHULE	374	21/06/2008	GALI NO.6, AZAD NAGAR, DHULE.	8080489594	MUSLIM	MUSLIM RANGARI	OPEN
45	P00CE2023-8618	23610960045	ROY JANHVI MANOHAR	FEMALE	02/08/2023	JAI HIND MADYAMIK VA UCHHA MADYAMIK VIDYALAYA CHALISGAON DIST JALGAON	347	21/04/2006	HARI OM NAGAR INFRONT OF NAGESHWAHAR MANDIR CHALISGAON	8421996603	HINDU	MOCHI	SC
46	P00CE2023-8590	23610960046	SAIFEE IDRIS MALIKULASHTER	MALE	31/07/2023	GARUD PRIMARY AND HIGH SCHOOL,DHULE.	359	05/03/2007	31,COTTON MARKET NEAR PUNJAB SAW MILL,DHULE	8394114116	MUSLIM		OPEN
47	P00CE2023-8737	23610960047	CHAUDHARI SAMIKSHA CHANDRASHEKHAR	FEMALE	16/08/2023	NORTH POINT SCHOOL	248	25/12/2006	AT POST FAGNE NEAR JAIN MANDIR TAL & DIST DHULE	8551049486	HINDU	TELI	OBC
48	P00CE2023-8803	23610960048	SHARMA PRATHA PRAMOD	FEMALE	25/08/2023	DHULE EDUCATION SOCIETY ENGLISH MEDIUM SCHOOL, DHULE.	372	23/10/2007	PLOT NO. 1, SHARMA NAGAR, NEAR NAVJIVAN BLOOD BANK, DHULE.		HINDU	HINDU MARWADI	OPEN
49	P00CE2023-8817	23610960049	SHARMA SHREYA SANDEEP	FEMALE	26/08/2023	CANOSSA CONVENT SCHOOL DHULE	346	24/11/2007	LANE NO 2 NEAR JAIN MANDIR DHULE	9422727559	HINDU	MARWADI	OPEN

Page 4 of 5

2. Literature Review



Maharashtra State Board of Technical Education, Mumbai

RESULT SHEET FOR THE FOURTH SEMESTER (K) EXAMINATION HELD IN SUMMER 2025 (INSTITUTE COPY)

INSTITUTE : 00059 - S. S. Vidya Prasarak Sanstha's Polytechnic COURSE : CO - Diploma in Computer Engineering

	1	2	3	4	5	6	7	8	9	10	11	12	13	14
	EES	EES	JPR	JPR	JPR	DCN	DCN	DCN	MIC	MIC	MIC	PWP	UID	UID
	TH	SLA	TH	PR	SLA	TH	PR	SLA	TH	PR	SLA	PR	PR	SLA
	100	025	100	075	025	100	050	025	100	050	025	100	050	025
	040	010	040	030	010	040	020	010	040	020	010	040	020	010

SEAT NO.	ENROLL NO	NAME	STATUS	APP.CODE	CF	Total Marks : 850
405664	24610960306	MARATHE LATIKESH SUBHASH	R	Y		

055	022	051	044	021	034	020	018	045	023	022	044	022	021
022		026	021		026	021		026	022		042	022	
077		077	065		060	041		071	045		086	044	

III Sem/W24/CO3K/379480/635/850/Add Marks :2/FailSub :0/Pending Sem :0 Total : 670 Result : FIRST CLASS WITH DISTINCTION TCALSE:60 / Credits:20

405665	24610960307	PADMAR PRAMOD DNYANESHWAR	R	Y									
025	018	028	034	019	020	013	014	016@	014	022	034	017	017
022		021	019		020	016		020	019		038	020	
047		049	053		040	029		036@	033		072	037	

III Sem/W24/CO3K/379481/476/850/Add Marks :0/FailSub :0/Pending Sem :0 Total : 486 Result : SECOND CLASS CON TCALSE:60 / Credits:20

405666	24610960308	PADVI MITESH AJABSING	R	Y									
032	022	035	032	015	011@	014	013	020	019	021	032	017	018
022		020	018		020	017		020	019		035	020	
054		055	050		031@	031		040	038		067	037	

III Sem/W24/CO3K/379482/431/850/Add Marks :0/FailSub :0/Pending Sem :0 Total : 492 Result : SECOND CLASS CON TCALSE:60 / Credits:20

Result Date : 20/06/2025 4/0059/CO4K

Page 446 of 494

Umesh T. Nagdeve
SECRETARY

- The above image represents the collected information of students used for the study. It includes important details such as student name, age, academic performance, and other relevant data required for analysis. This information helps in understanding the characteristics and behavior patterns of students. The data collected from students plays a significant role in evaluating the effectiveness of the system and supports the findings of the research.

3. Scope of the project

A. User Roles and Modules:

Student:- Student Registration and Login.

- Profile Creation and Management.
- Upload Photo and Academic Details.
- Submit Change Request.

Teacher:- Teacher login.

- View Pending Students Profile.
- Verify and Reject Students Profile.
- Manage and Change Request.

Admin :- Add and manage teacher accounts.

- Provide system access to teachers.
- Monitor student contact information.
- Manage and maintain the website.

4. Methodology

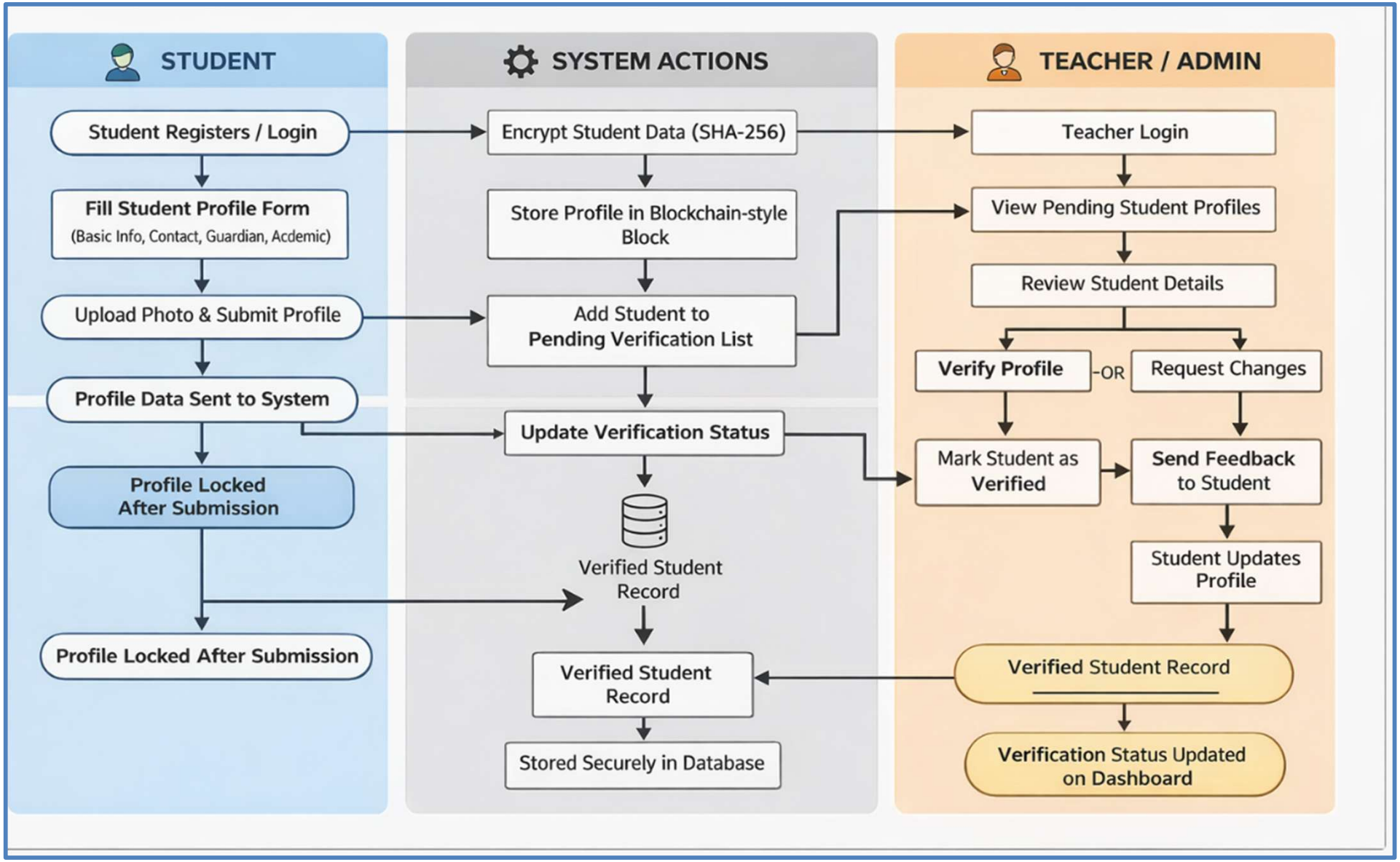
B. Functional Scope: Student Profile management.

- Secure encrypted record storage.
- Teacher verification system.
- Role based Access control.

- **In future:-**

- Integration with external education databases No integrated payment gateway
- Biometric authentication system.
- Full decentralized blockchain implementation.

5. Details of designs, working & processes



5. Details of designs, working & processes

A. **Student Working:-** The student registers or logs into the system and fills out the profile form with personal, contact, guardian, and academic details. After uploading a photo and submitting the form, the system encrypts the data and stores it securely using a blockchain-inspired method. The profile becomes locked after submission. The student can check the verification status and update the profile if the teacher requests changes.

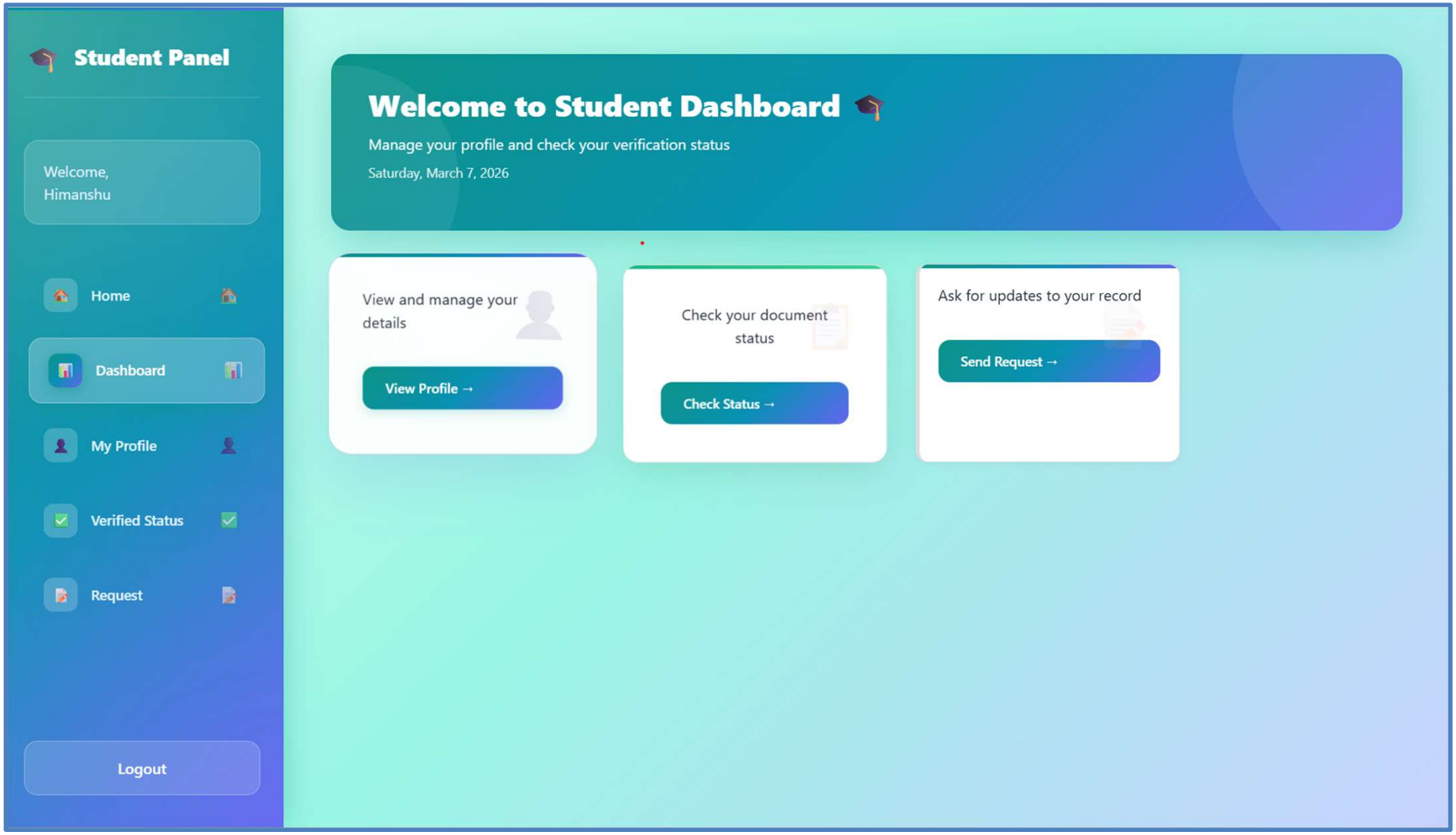
B. **Teacher Working:-**The teacher logs into the system and views the list of students whose profiles are pending verification. The teacher reviews the student details and either verifies the profile or requests changes if needed. Once verified, the student's record is marked as verified in the system.

5. Details of designs, working & processes

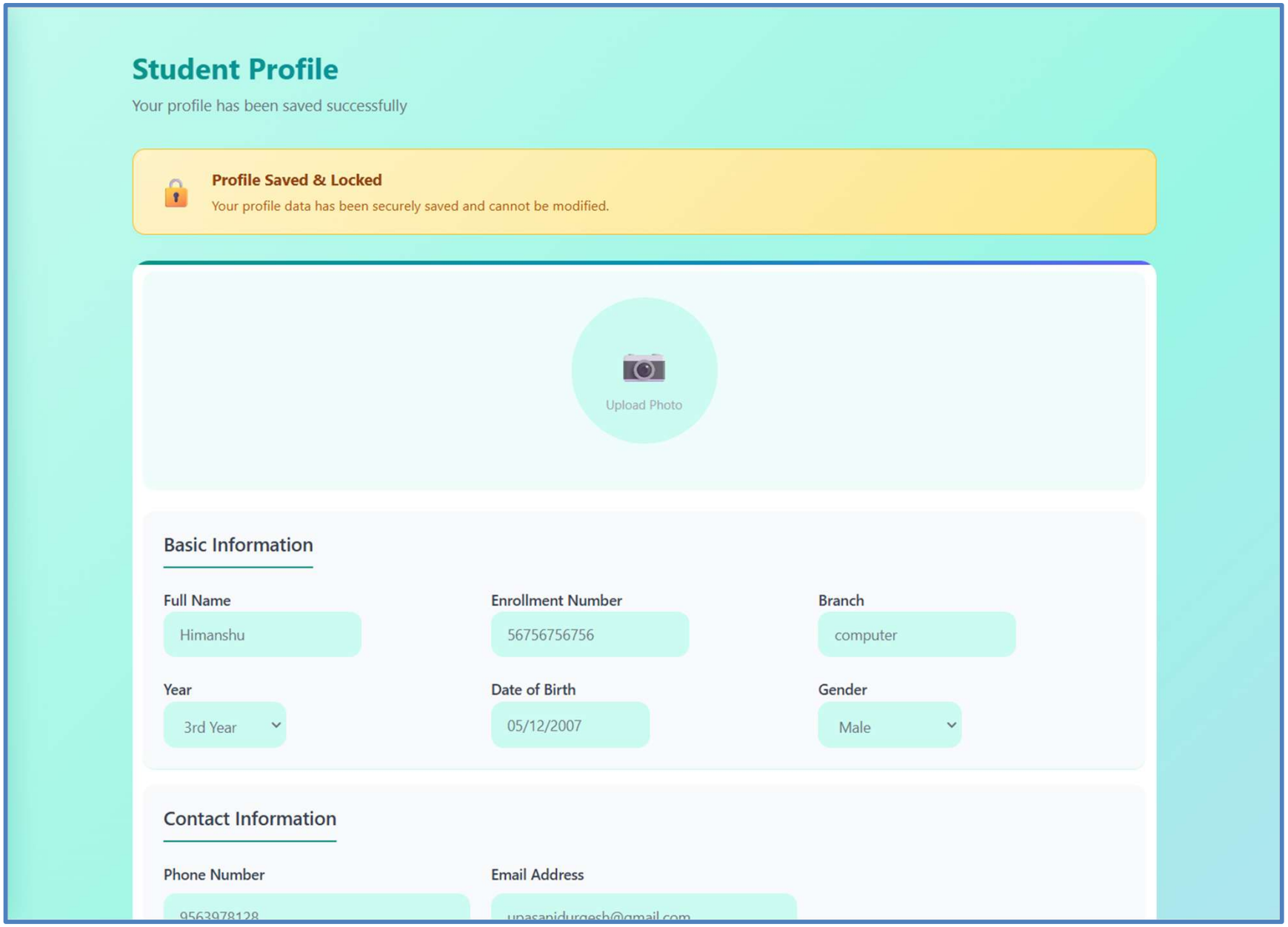
C. System Automated Actions:- The system manages the overall process of storing and verifying student records. It encrypts student profile data using SHA-256 and stores it securely in a blockchain-style structure. The system updates verification status, manages requests, and ensures secure access through authentication.

D. Admin Working:-The admin logs into the system and manages the overall platform. The admin can add and manage teacher accounts, provide system access to teachers, monitor student information and requests, and ensure the system operates smoothly.

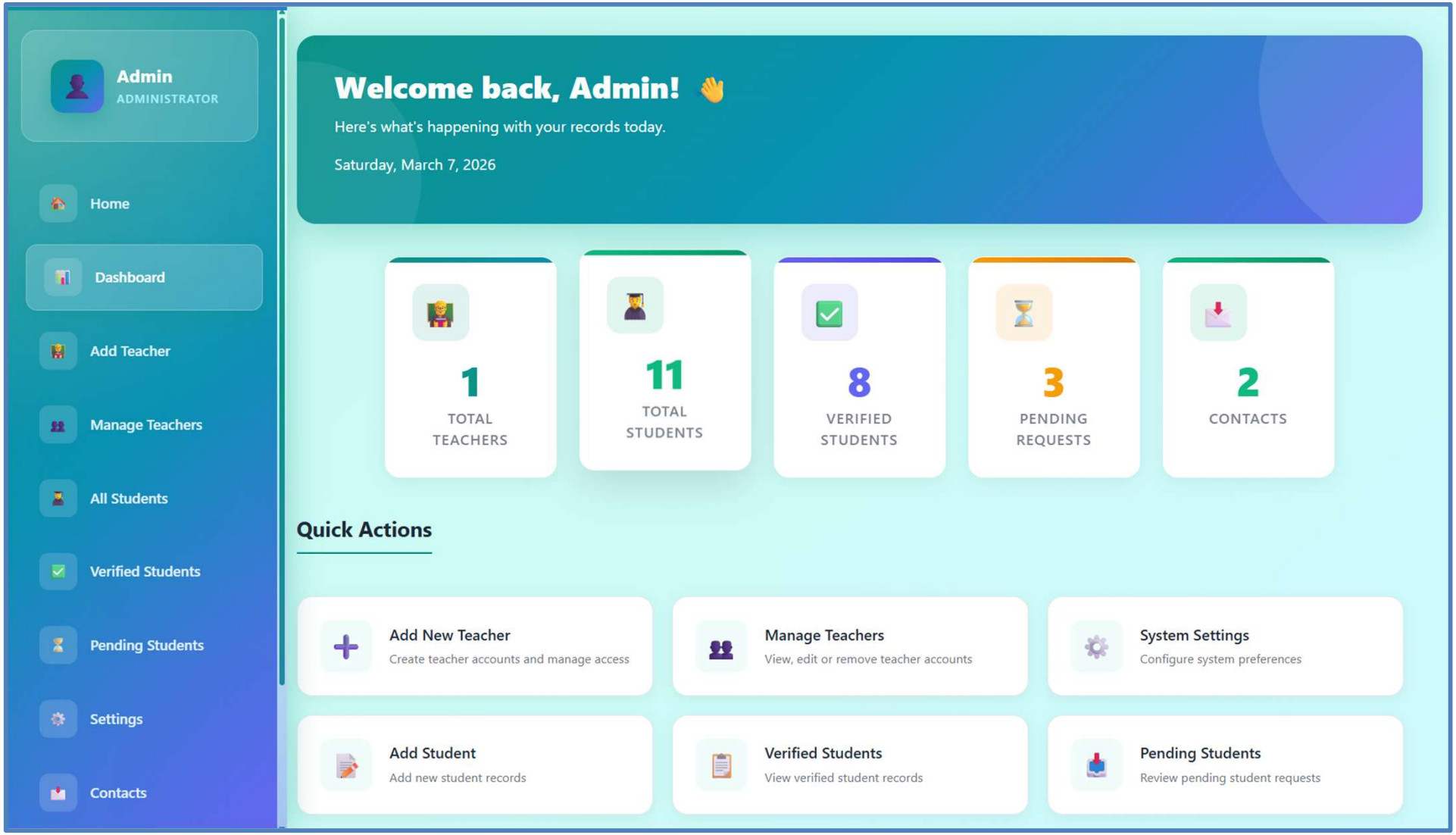
6. Results and Applications



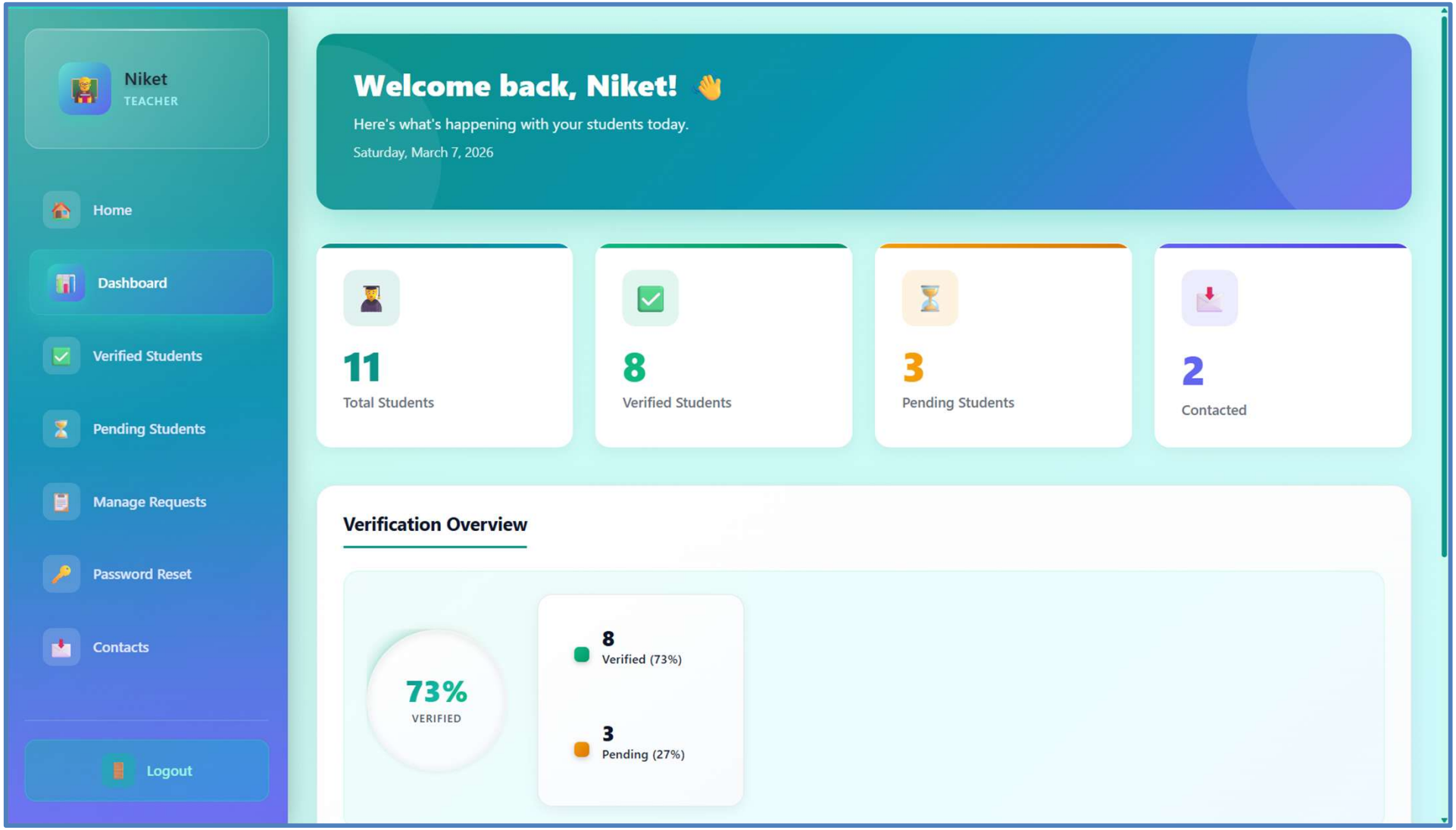
6. Results and Applications



6. Results and Applications



6. Results and Applications



7. Conclusion and future scope

- **Conclusion:-**

This project presents a secure and efficient student record management system using blockchain technology. By storing cryptographic hashes of academic records on the blockchain, the system prevents data tampering and eliminates fake certificates. It allows students to access their records easily, while universities and administrators can verify them quickly and reliably. The system improves security, transparency, and trust in academic record management. Overall, it provides a practical and reliable solution that is suitable for real-world educational applications.

7. Conclusion and future scope

- **Future Scope:-**

•Integration with real blockchain technology.

•Email and SMS notification system.

•Development of a mobile application.

•Advanced analytics and reporting features.

•Integration with government or educational databases.

•Two-factor authentication for better security.

8. References and Bibliography

- **References:**

1. https://github.com/
2. https://www.geeksforgeeks.org/
3. https://www.ibm.com/topics/blockchain

- **Bibliography:-**

1. **Chen, G., Xu, B., Lu, M., & Chen, N. (2018).**

➢ Explores how blockchain improves transparency and trust in educational systems.

2. **Crosby, M., Pattanayak, P., Verma, S., & Kalyanaraman, V. (2016).**

➢ Describes blockchain architecture, security, and real-world applications.

3. **Zyskind, G., Nathan, O., & Pentland, A. (2015).**

➢ Explains blockchain-based secure data management systems.